BASIC SORTING SOLUTIONS

Solution (a): Bubble Sort

```java
public static void bubbleSortDescending(int arr[]) {
        for(int turn=0; turn<arr.length-1; turn++) {
            for(int j=0; j<arr.length-1-turn; j++) {
                if(arr[j] < arr[j+1]) {
                    //swap
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
    }
```

Solution (b): Selection Sort

```java
public static void selectionSortDescending(int arr[]) {
        for(int turn=0; turn<arr.length; turn++) {
            int minPos = turn;
            for(int j=turn+1; j<arr.length; j++) {
                if(arr[minPos] < arr[j]) {
                    minPos = j;
                }
            }

            //swap
            int temp = arr[turn];
            arr[turn] = arr[minPos];
            arr[minPos] = temp;
        }
    }
```

Solution (c) : Insertion Sort

```java
public static void insertionSortDescending(int arr[]) {
```

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```
        for(int i=1; i<arr.length; i++) {
            int curr = arr[i];
            int prev = i-1;
            //to find the index where curr is to be inserted
            while(prev >= 0 && arr[prev] < curr) {
                arr[prev+1] = arr[prev];
                prev--;
            }
            arr[prev+1] = curr;
        }
    }
```

Solution (d): Counting Sort

```
public static void countingSortDescending(int arr[]) {
        int largest = Integer.MIN_VALUE;
        for(int i=0; i<arr.length; i++) {
            largest = Math.max(largest, arr[i]);
        }

        int count[] = new int[largest+1];
        for(int i=0; i<arr.length; i++) {
            count[arr[i]]++;
        }
        int j = 0;
        for(int i=count.length-1; i>=0; i--) {
            while(count[i] > 0) {
                arr[j] = i;
                j++;
                count[i]--;
            }
        }
    }
```

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